

Kundai Farai Sachikonye

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🌐 github.com/fullscreen-triangle

*Computational scientist and ML engineer: explainable,
multimodal AI over clinical text, imaging, and laboratory data —
from method to deployed, reproducible, reference-validated
system.*

Profile

Computational scientist and machine-learning engineer working on explainable, multimodal AI for biomedicine. I build deep-learning and multi-omics pipelines over clinical and laboratory data, GPU-accelerated medical image analysis, and auditable diagnostic models — and take each from method to deployed, documented, reference-validated tool. I have worked on clinical data inside two university hospitals and conducted doctoral research at TUM. I am seeking a structured doctorate to ground this work in a clinical, interdisciplinary setting.

Education

- 2019–2021 **Doctoral research, Computational Lipidomics**, *Technische Universität München*, Munich
ML for large-scale clinical omics cohorts; distributed inference across federated clinical sites; HPC-scale statistical analysis. Taught at Master's level (in German). Programme not completed (funding ended); research continued independently.
- 2018–2019 **MSc Computational Biology**, *Universität Tübingen*
Probabilistic graphical models, statistical bioinformatics, Bayesian methods, ML for biological sequences. **Conducted entirely in German.**
- 2014–2017 **MSc Pharmaceutical Biotechnology**, *HAW Hamburg*
Conducted entirely in German.
- 2011–2014 **BSc Biochemistry and Cell Biology**, *Jacobs University Bremen*
Fluorescence-microscopy thesis (DNA interstrand crosslinking).

Technical Skills

Machine learning	PyTorch (CNNs, attention / transformers, autoencoders); representation learning; foundation-model adaptation (LoRA), knowledge distillation; scikit-learn, XGBoost; model evaluation & benchmarking
Multimodal & imaging	GPU medical-image analysis — segmentation, deconvolution, registration, morphometry; camera-based (RGB) physiological signal recovery; fusion of physically distinct signal channels into one inference; uncertainty quantification
Explainable / auditable AI	Zero-parameter, first-principles diagnostic models reproducible from the algorithm alone; structural audibility vs. post-hoc explanation (SHAP / LIME)
Data & infrastructure	HPC / SLURM, Ray, Dask; Docker, Kubernetes; FastAPI; HDF5, Parquet, memory-mapped I/O for >100 GB datasets; reproducible, seeded, benchmarked pipelines
Clinical / omics data	LC-MS/MS & MALDI-TOF proteomics; multi-omics integration; variant analysis; reference data — 1000 Genomes, gnomAD, UK Biobank, IEDB, NIST
Programming	Python (primary), Rust, TypeScript, R, SQL, Bash; GLSL / WebGPU

Clinical & Research Experience

- 2021–present **Independent Computational Science & ML Researcher**, *Bitspark GmbH*
 Deep-learning and multi-omics pipelines, GPU image analysis, agentic/multi-agent systems, and auditable diagnostic tools — from concept to deployed, documented, reference-validated software. All work open-source.
- 2019–2021 **Computational Science — ML & HPC**, *Technische Universität München*, Munich
 Deep-learning pipelines (PyTorch, CNNs) at HPC scale; SLURM; statistical evaluation of >100 GB clinical-omics datasets. Master’s-level teaching, in German.
- 2017–2018 **Glycomics & Proteomics**, *Universitätsklinikum Hamburg-Eppendorf*, Hamburg
 Automated LC-MS/MS and MALDI-TOF clinical data pipelines; reproducible workflows in a university-hospital setting.
- 2013 **Cancer Biomarker Discovery**, *University Medical Center Groningen*, Groningen
 UPLC-MS/MS proteomics; high-dimensional feature analysis for oncology biomarkers.

Selected Projects (open-source, deployed)

- syndrome **Explainable oncology AI**. Neoantigen identification, MHC binding (AUC 0.81, 2,847 IEDB peptides), and immune-escape scoring via 3-channel DFT on amino-acid properties — zero parameters trained on patient data, every prediction reproducible from the algorithm alone. Developed with KRAS-mutant pancreatic (PDAC) immune evasion as a worked case. syndrome-seven.vercel.app
- hieronymus **GPU medical image analysis**. Segmentation, point-spread-function deconvolution, registration, and morphometry for microscopy; WebGL/GLSL; metric-preserving reconstruction with uncertainty. hieronymus-omega.vercel.app
- BRUT (rPPG) **Camera-based physiological inference**. Recovers melanin, haemoglobin, and vasodilation from RGB video by inverting a layered optical skin model — multimodal signal fusion from a single imaging stream. remote-photoplethysmography.vercel.app
- lavoisier **Reproducible laboratory-data pipeline**. High-throughput mass-spectrometry processing; 98.9% NIST accuracy; distributed (Ray); deterministic and benchmarked. lavoisier.vercel.app/sandbox
- gospel **Multi-site multi-omics integration**. Federated inference (genotype + transcriptomics + metagenomics) as a joint Bayesian graph; validated against 1000 Genomes, gnomAD, UK Biobank. github.com/fullscreen-triangle/gospel
- kwasa-kwasa, four-sided-triangle, purpose **Agentic AI & reasoning**. A multi-agent orchestration language (Rust + TypeScript, with a language server); an 8-stage metacognitive RAG/reasoning pipeline (FastAPI + Ray + Docker); and a context-retrieval CLI that cuts LLM token burn. github.com/fullscreen-triangle

Languages

- English Native / C2
- German Two master’s degrees completed entirely in German; taught master’s-level courses in German at TUM
resident
in
Ger-
many
since
2011